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Big Data Analytics Project

NYC Flight Delays + Weather Data

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Install packages once

install.packages(c(“tidyverse”,“nycflights13”,“lubridate”,“cluster’

library(tidyverse) library(nycflights13) library(lubridate) library(cluster) library(arules) library(arulesViz) library(ggplot2)

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1. Load & Merge Datasets

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df <- flights %>% select(year, month, day, dep_delay, arr_delay, carrier, air_time, distance, origin, time_hour)

Add weather (external factor)

df <- df %>% left_join(weather, by=“time_hour”)

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2. Data Cleaning

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df <- df %>% drop_na(dep_delay, arr_delay, air_time, distance, temp, wind_speed, visib)

Convert carrier to factor

df$carrier < -as.factor(df$carrier)

Create derived variable: Delay Category

df <- df %>% mutate(delay_cat = case_when(arr_delay <= 0 ~ “On Time”, arr_delay > 0 & arr_delay <= 30 ~ “Minor Delay”, arr_delay > 30 ~ “Major Delay”))

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3. Regression Analysis:

Predict arrival delay

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reg_model <- lm(arr_delay ~ dep_delay + air_time + distance + temp + wind_speed + visib + carrier, data=df)

reg_output <- summary(reg_model) reg_output

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4. Clustering

cluster_data <- df %>% select(dep_delay, arr_delay, air_time, distance, temp, wind_speed, visib) %>% scale()

set.seed(123) k <- kmeans(cluster_data, centers=3) df$cluster < -as.factor(k$cluster)

Cluster Plot

ggplot(df, aes(arr_delay, dep_delay, color=cluster)) + geom_point(alpha=0.3) + theme_minimal() + labs(title="Flight Delay Clusters (k=3)")